

- 8 The driver of a 1000 kg car tries to turn through a circle of radius 100 m on an unbanked curve at a speed of 10 m s^{-1} . The maximum frictional force between the tyres and the slippery road is 900 N. The car will
- A Slide into the inside of the curve.
 - B Make the turn.
 - C Slow down due to the centrifugal force.
 - D Slide off to the outside of the curve.

Ans: D

Centripetal force required, $F_c = mv^2/r = 1000 \times (10^2)/100 = 1000 \text{ N}$

The maximum frictional force between the tyres and the slippery road is 900 N, which is insufficient to allow the car to turn around. Hence, the car will slide off to the outside of the curve.

- 9 A simple pendulum consists of a bob of mass m at the end of a light and inextensible thread of